

PAVAN KUMAR KUNASANI

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PROFESSIONAL PROFILE

Senior Full-Stack Engineer with **11+ years of experience** delivering scalable enterprise applications across the complete Software Development Lifecycle. Specialized in **Java, Spring Boot 3, and Microservices architecture**, with strong expertise in **System Design, distributed transactions, and high-performance backend development**.

Experienced in building responsive UI solutions using React and in modern containerized deployments using **Docker/Podman on Microsoft Azure cloud**. Proven ability to **lead technical initiatives, optimize system performance, and deliver resilient, production-ready solutions**. Seeking a **Technical Lead role** in driving architecture decisions and building scalable digital platforms.

TECHNICAL EXPERTISE

Architecture & Design

System Design, Microservices Architecture, Distributed Transactions, Object-Oriented Design (OOD), Design Patterns

Backend Development

Java (8/11+), Spring Boot 3, Spring MVC, Spring Security, Hibernate/JPA, RESTful API Design

Messaging, Caching & Data

RabbitMQ, Redis (Distributed Caching), Oracle 11g, SQL, MongoDB (NoSQL)

Frontend Technologies

React, JavaScript, HTML5, CSS3

Cloud & DevOps

Microsoft Azure, Docker, Podman, Jenkins CI/CD, Maven, Git, SVN

Testing & Supporting Tools

JUnit, Mockito, Log4j, Apache Tomcat, JBoss, Postman, DBeaver

PROFESSIONAL EXPERIENCE

Senior Software Engineer | Concentrix | December 2025 – Present

Projects: CPSL (Crew Pro Short Lines)

CrewPro Short Lines (CPSL) is an enterprise application designed for the **railroad domain**, enabling efficient management of **crew operations, task scheduling, and workforce administration**. The system supports **daily crew assignments, future task planning, leave management, and payroll processing**, helping organizations streamline operational workflows and improve workforce visibility.

Key Contributions:

- Led migration of monolithic modules into **Spring Boot microservices**, enabling independent deployment and scalability.
- Designed **RESTful APIs and service boundaries** for crew scheduling and payroll workflows.
- Collaborated with cross-functional teams to deliver **containerized, cloud-ready services**.
- Identified and resolved **performance bottlenecks** in legacy systems during modernization.
- Contributed to **architecture and design discussions** for scalable system evolution.

Senior Software Engineer | Concentrix | January 2019 – November 2025

Projects: QUAL PRO - Qualification Tracking Management

Qual Pro is an enterprise application focused on **compliance-driven qualification management**, enabling organizations to track, manage, and validate workforce certifications and regulatory requirements. The system supports **high-volume transactional workflows, data-intensive operations, and real-time visibility into qualification status**, ensuring adherence to industry compliance standards.

Key Contributions:

- Modernized legacy modules into **Spring Boot-based services**, improving scalability and maintainability.
- Collaborated with clients and product stakeholders to **gather requirements and define user stories**, aligning technical solutions with business needs.
- Performed **impact analysis for new features and enhancements**, minimizing disruption to existing workflows.
- Designed and developed **data-intensive Angular components**, improving usability and compliance with data visibility.
- Optimized **database queries and ORM mappings**, ensuring efficient processing of high-volume transactions.
- Managed **CI/CD pipelines using Jenkins**, enabling automated build and deployment processes.

- Deployed and supported applications on **Microsoft Azure**, ensuring high availability and scalable infrastructure.

Software Engineer | HTC Global Services | March 2018 – January 2019

Project: ICMA - Specialized Nursing Homes Program

ICMA is a healthcare-focused application designed to support **specialized nursing home programs**, enabling patients and providers to manage services, access program information, and streamline healthcare workflows through a digital platform.

Key Contributions

- Designed and developed a **customer self-service module** using Spring Boot and Hibernate, improving accessibility to healthcare program workflows.
- Built and documented **RESTful APIs using Spring MVC and Swagger**, enabling seamless integration with external client systems.
- Implemented **unit and integration testing** using JUnit and Mockito, improving reliability of critical healthcare services.
- Contributed to backend service development and **data persistence layer design**, ensuring consistency and maintainability.

Software Engineer | Verinon Technology Solutions Pvt. Ltd. | January 2015 – March 2018

Project: LISI Systems

LISI Systems is an enterprise application supporting the aerospace supply chain, focusing on shipping, logistics, and compliance management to meet strict operational and regulatory standards.

Key Contributions:

- Developed and enhanced **shipping and logistics modules**, ensuring compliance with aerospace operational standards.
- Implemented backend workflows using **Spring MVC and IoC-based architecture**, improving system modularity, and maintainability.
- Provided **production support and issue resolution** for critical supply chain operations, ensuring system reliability.
- Collaborated with business analysts to **translate requirements into technical solutions** for evolving logistics processes.

Project: Doc-Viewer

Overview:

Doc-Viewer is an internal tool designed to **view, manage, and process enterprise documents**, improving accessibility and document handling efficiency.

Key Contributions:

- Developed document handling features using **Java-based web technologies**, enabling efficient document viewing and processing.
- Improved system usability by enhancing **UI interactions and data presentation**.
- Assisted in debugging and resolving **application-level issues** during development and support phases.

Project: Annual Assessment

Overview:

Annual Assessment is an HR-focused application used to **manage employee performance evaluations and appraisal workflows**.

Key Contributions:

- Implemented core modules for **employee assessment and evaluation workflows** using Spring MVC.
- Worked with stakeholders to **enhance features based on business requirements**.
- Supported application maintenance and ensured **smooth execution of appraisal cycles**.

Project: Retrieval-Augmented Knowledge Search System (Personal Project)

Overview:

Developed a **Retrieval-Augmented Generation (RAG) system** for semantic knowledge search, enabling context-aware query handling over unstructured documents. The system uses **Hugging Face embeddings models** for vector representation and **MongoDB vector database** for storage and retrieval.

- Designed and implemented an **end-to-end RAG pipeline**, combining semantic retrieval with LLM-based response generation.
- Built **document ingestion and indexing workflows** to process PDFs and text files into vector representations for efficient semantic search.
- Developed **RESTful APIs** to handle user queries, integrating the retrieval and response generation layers.
- Implemented **semantic search** using vector embeddings, improving relevance and contextual understanding over traditional keyword-based search.
- Optimized query processing and retrieval to handle **high-volume document searches** efficiently.

Awards & Recognition

- Received Performance Excellence Award at Concentrix on three occasions in recognition of consistent delivery, system modernization contributions, and performance optimization initiatives.
- Acknowledged by leadership for driving scalable microservices adoption and improving enterprise application stability.

EDUCATION

Master's Degree (Pursuing)

Masters in computer science

Bachelor of Technology (Electronics & Communication Engineering)

Jawaharlal Nehru Technological University, Kakinada, India